

## STAFFORD 3.8: TECHNICAL FORMAL-PROOF COMPARISONS

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**AI comment Technical review. Information — readable proof accounts.**

This supplement explains the formal arguments as mathematics. These accounts are AI additions for author review. References point to the human-readable paper; this supplement is not required reading for editing that paper.

### Comparison COMP-01

**AI comment COMP-01. Information — no paper edit requested.** Lean

follows the printed localization proof: extend the bracket by the quotient rule, use minimality to identify the localized radical with the maximal ideal, and contract. Its bracket has the opposite sign; involutivity is unchanged. [Mathematical proof \(COMP-01, p. 1\)](#).

#### AI addition: Proof as formalized in Lean.

Lean proves the lemma in the corrected form of ALG-01. Let  $J$  be any ideal of  $R = k[x, \xi]$  with  $\{\sqrt{J}, \sqrt{J}\} \subseteq \sqrt{J}$ , and let  $\mathfrak{p}$  be minimal over  $J$ . Then  $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$ . The official proof follows the paper’s localization argument. The partial derivatives extend to  $R_{\mathfrak{p}}$  by the quotient rule

$$\partial_i(a/s) = (s\partial_i a - a\partial_i s)/s^2.$$

Their Poisson bracket is a biderivation and preserves the extension of  $\sqrt{J}$ . Minimality gives  $(\sqrt{J})R_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}$ , so this maximal ideal is involutive. Contracting it to  $R$  proves the assertion. Lean’s bracket has the opposite sign from the displayed paper convention; involutivity is unchanged.

The official Gabber interface consumes the localized proof above to pass from radical involutivity to minimal-prime involutivity.

### Comparison COMP-02

**AI comment COMP-02. Information — no paper edit requested.** Lean

checks the printed induction in an arbitrary symplectic space, including completion with the prescribed first vector. The official normalization uses the resulting basis-to-basis map and its matrix. [Mathematical proof \(COMP-02, p. 1\)](#).

#### AI addition: Proof as formalized in Lean.

Lean follows the printed basis induction in an arbitrary finite-dimensional space with an alternating nondegenerate form. For a prescribed nonzero  $v$ , choose  $w$  with  $\omega(v, w) = 1$ . The plane  $U = \text{span}(v, w)$  is nondegenerate; Lean proves  $\mathbb{V} = U \oplus U^{\perp}$ , nondegeneracy on  $U^{\perp}$ , and its dimension  $2n - 2$ .

Induction gives a symplectic basis of the complement and hence a basis beginning with  $v, w$ .

Completing each of two nonzero vectors gives bases with the same pairwise form values. The basis-to-basis map therefore preserves the form and sends one prescribed vector to the other. The official normalization caller converts this map into its matrix and checks the standard-form identity. The abstract basis pairing and the coordinate matrix use opposite Gram sign conventions; the conversion proves the matrix identity explicitly. No change to the paper's basis proof is needed.

### Comparison COMP-03

**AI comment COMP-03. Information — proof comparison; no edit requested.** The formal normalization uses an explicit symplectic matrix and its inverse. Its symbol substitution is  $s(MX)$ ; with the convention used here this is  $s \circ g^\top$ . Monicity means that the  $p^N$  PBW coefficient is 1 and the Bernstein degree is at most  $N$ , which gives the coefficient bounds above. [Mathematical proof \(COMP-03, p. 2\).](#)

### AI addition: Proof as formalized in Lean.

Lean writes the change of generators with a matrix  $M = g^\top$ :  $\Phi_M(z_i) = \sum_j M_{ij} z_j$ , where  $z = (x_1, \dots, x_n, \partial_1, \dots, \partial_n)$ . Put  $J_{ij} := [z_i, z_j]$ . Then  $[\Phi_M z_i, \Phi_M z_j] = (M J M^\top)_{ij}$ , so  $\Phi_M$  is an algebra endomorphism whenever  $M J M^\top = J$ . If  $M'$  also satisfies this and  $M M' = M' M = 1$ , then  $\Phi_{M'}$  is the inverse of  $\Phi_M$ . For  $a \in B_N$  one has  $\Phi_M(a) \in B_N$  and  $\sigma_N^B(\Phi_M a)(X) = \sigma_N^B(a)(MX)$ , where  $(MX)_i = \sum_j M_{ij} X_j$ .

Let  $N$  be the Bernstein degree of  $d$ , i.e. the top total degree of its PBW normal form. Let  $s = \sigma_N^B(d) \neq 0$ ; it is homogeneous of degree  $N$ . If  $N = 0$ ,  $d$  is a nonzero scalar. Let  $N \geq 1$  and let  $t$  be the index of  $\partial_1$ . The argument has five steps.

(i) Since  $k$  is infinite, there is  $v \in k^{2n}$  with  $s(v) \neq 0$ . Here  $v \neq 0$ , because  $s(0) = 0$  for  $N \geq 1$ .

(ii) Lemma 4.1 with  $e = e_t$  gives a symplectic  $M$  whose  $t$ -th column is  $v$ , i.e.  $M e_t = v$ . Its inverse  $M'$  is symplectic too.

(iii) The symbol  $s_M(X) := s(MX)$  of  $\Phi_M(d)$  is homogeneous of degree  $N$ . On the  $\xi_1$ -axis it restricts to  $c \xi_1^N$ , where  $c$  is its coefficient of  $\xi_1^N$ . Hence  $c = s_M(e_t) = s(M e_t) = s(v) \neq 0$ .

(iv) Consequently  $\tilde{d} := c^{-1} \Phi_M(d)$  lies in  $B_N$ , and the coefficient of  $\xi_1^N$  in  $\sigma_N^B(\tilde{d})$  is 1. This records the chosen symplectic change of coordinates together with its normalized leading coefficient.

(v)  $\sigma_N^B(\tilde{d})$  is the weight- $N$  part of the PBW normal form, so this coefficient is the coefficient of the PBW word  $\partial_1^N$  in  $\tilde{d}$ . The resulting normal form has these properties: every PBW word  $x'^\alpha \partial'^\beta x_1^a p^b$  of  $\tilde{d}$  has  $|\alpha| + |\beta| + a + b \leq N$ , and  $p^N$  has coefficient 1. The words are ordered transverse pairs first, then  $x_1^a p^b$ , with  $p$  rightmost. Grouping by  $b$  gives  $\tilde{d} = p^N + \sum_{j < N} c_j p^j$  with  $c_j \in C$

and  $\deg_B c_j \leq N - j$ , with the coefficients on the *left*. Lean records this as monicity of the outer Ore polynomial.

Lean additionally checks the paper's right-coefficient form  $\tilde{d} = p^N + \sum_{j < N} p^j a_j$ . Moving coefficients across  $p$  uses the coefficient derivation, which lowers Bernstein degree. The right PBW basis therefore gives  $\deg_B a_j \leq N - j$ , top coefficient  $a_N = 1$ , and zero coefficients above  $N$ . Its finite-support reconstruction proves the stated equality. The paper can keep its original right-coefficient convention.

### Comparison COMP-04

**AI comment COMP-04. Information — no paper edit requested.** Lean checks the Euler weights of PBW monomials and identifies the nonnegative Euler part with the transverse–Euler subring used in the residue argument. The corrected product identities and one-sided normality give the same surjectivity conclusion. [Mathematical proof \(COMP-04, p. 2\).](#)

#### AI addition: Proof as formalized in Lean.

Lean checks the Euler weights of PBW monomials directly:  $[E, bx_1^a p^j] = (a - j)bx_1^a p^j$  for transverse  $b$ . It proves that the subring generated by the nonnegative-weight PBW monomials is exactly the subring  $R \subseteq A_n$  generated by  $T' = A_{n-1}(k)$ ,  $x_1$  and  $E = x_1 p$ . This identifies the paper's nonnegative Euler part with the subring used by the residue proof. Internally, Lean's Ore normal form puts the coefficients on the left: every element is uniquely  $\sum_j c_j p^j$  with  $c_j \in C = T'[x_1]$ , that is, a sum of words  $bx_1^a p^j$  with  $b \in T'$ . The proof has five steps.

(a)  $R$  contains  $\mathbb{Q}[E]$ . It is *one-sidedly normal*: for  $r \in R$  there is  $r' \in R$  with  $x_1 r = r' x_1$ . The set of such  $r$  is a subring, and it contains the generators, because  $x_1 b = bx_1$ ,  $x_1 x_1 = x_1 x_1$  and  $x_1 E = (E - 1)x_1$ .

(b) The polynomials  $\prod_{i=0}^{N-1} (X - i)$  and  $\prod_{i=1}^N (X + i)$  have the disjoint root sets  $\{0, \dots, N - 1\}$  and  $\{-1, \dots, -N\}$ . They are therefore coprime in  $\mathbb{Q}[X]$ . With Lemma 5.1 (corrected form) this gives  $u, v \in \mathbb{Q}[X]$  with  $p^N x_1^N u(E) + x_1^N p^N v(E) = 1$ .

(c) The normal form of Proposition 4.2 gives  $\tilde{d} = p^N + \sigma$ , where  $\sigma$  is a sum of words  $bx_1^a p^j$  with  $j < N$ . Then

$$\tilde{d}(x_1^N u(E)) + (x_1^N \tilde{d})v(E) = 1 + \sigma x_1^N u(E) + x_1^N \sigma v(E),$$

and the left side lies in  $I$ .

(d) Each word  $w = bx_1^a p^j$  with  $j < N$  has a right factor  $x_1$  on both sides, with cofactors in  $R$ . By Lemma 5.1 and (a),

$$\begin{aligned} w x_1^N &= b x_1^a (p^j x_1^j) x_1^{N-j} = \left[ b x_1^a \prod_{i=1}^j (E + i) x_1^{N-j-1} \right] x_1, \\ x_1^N w &= b x_1^{N+a-j-1} x_1 (x_1^j p^j) = \left[ b x_1^{N+a-j-1} \prod_{i=0}^{j-1} (E - 1 - i) \right] x_1. \end{aligned}$$

By additivity,  $\sigma x_1^N = U'x_1$  and  $x_1^N \sigma = V'x_1$  with  $U', V' \in R$ .

(e) Since  $x_1 u(E) = u(E-1)x_1$ , the residue equals  $Ux_1$  with  $U = U'u(E-1) + V'v(E-1) \in R$ . Hence  $1 + Ux_1 \in I$ .

### Comparison COMP-05

**AI comment COMP-05. Information — proof comparison; no edit requested.** The formal proof works element by element with the transverse-Euler subring. The residue identity gives  $1 + Ux_1 \in I$  for a suitable  $U$  in that subring; one-sided normality and the monic relation then give surjectivity of right multiplication by  $x_1$ . This avoids treating a quotient by a vector subspace as an algebra quotient. [Mathematical proof \(COMP-05, p. 3\).](#)

#### AI addition: Proof as formalized in Lean.

Lean argues element by element instead of through the submodule  $qA_{\geq 0}$ . Let  $R, U$  be as in Lemma 5.2 (Lean form), and write  $a \equiv b$  for  $a - b \in I$ . Put

$$\mathcal{P} := \{a \in A_n : \text{for every } r \in R \text{ there is } r' \in R \text{ with } ra \equiv r'\}.$$

$\mathcal{P}$  is a subring containing  $R$ . It is closed under products: if  $ra \equiv s$  and  $sb \equiv t$ , then  $rab \equiv sb \equiv t$ , because  $I$  is a right ideal. It also contains  $p$ . Indeed, for  $r \in R$  pick  $r' \in R$  with  $x_1 r = r'x_1$ . From  $(1 + Ux_1)rp \in I$  we get  $rp \equiv -Ux_1rp = -Ur'E \in R$ .

The Ore normal form shows that  $T', x_1$  and  $p$  generate  $A_n$ , and  $T', x_1 \in R$ . Hence  $\mathcal{P} = A_n$ . Given  $a \in A_n$ , take  $r = 1$ : then  $a \equiv r_a$  for some  $r_a \in R$ . Choosing  $r'$  with  $x_1 r_a = r'x_1$  gives  $a \equiv r_a \equiv -Ux_1 r_a = (-Ur')x_1$ . So the class of  $a$  is the class of  $-Ur'$  multiplied on the right by  $x_1$ .

Lean proves this in the iterated Ore model of  $A_n$  and transports it along the isomorphism with the presented Weyl algebra.

### Comparison COMP-06

#### AI addition: Proof as formalized in Lean.

Lean never localizes  $Q$ ; only the pages carry a  $T$ -module structure. Put  $G^p := F_{-p}Q$  for  $p \leq 0$  and  $G^p := 0$  for  $p > 0$ . This is a decreasing, exhaustive filtration with  $G^1 = 0$ . The complex is the  $k$ -linear map  $f = R_{x_1}: Q \rightarrow Q$ , with  $f(G^p) \subseteq G^p$ . For  $r \geq 0$  define

$$A_r^p = \frac{G^p \cap f^{-1}(G^{p+r})}{G^{p+1} \cap f^{-1}(G^{p+r})}, \quad B_r^p = \frac{G^p}{(G^p \cap f(G^{p-r+1})) + G^{p+1}},$$

and  $A_r = \bigoplus_{p \in \mathbb{Z}} A_r^p$ ,  $B_r = \bigoplus_{p \in \mathbb{Z}} B_r^p$ . Applying  $f$  induces  $d_r: A_r^p \rightarrow B_r^{p+r}$ : changing a representative by the source denominator changes its image by the target denominator. Thus  $A_0 = B_0 = M$  and  $d_0$  is multiplication by  $x_1$ . The proof has four steps.

*Step 1:  $T$  acts on the pages.* Let  $a$  be a transverse generator  $x_j$  or  $\partial_j$  ( $j \geq 2$ ), of order  $e_a \in \{0, 1\}$ . Then  $R_a$  is  $k$ -linear,  $R_a(G^p) \subseteq G^{p-e_a}$ , and  $R_a$  commutes with  $f$  because  $ax_1 = x_1a$ . It therefore induces maps  $A_r^p \rightarrow A_r^{p-e_a}$  and  $B_r^p \rightarrow B_r^{p-e_a}$  that commute with  $d_r$ . For two generators,  $[R_a, R_b] = \pm R_{[a,b]}$  with  $[a,b] \in F_{e_a+e_b-1}$ , so it maps  $G^p$  into  $G^{p-e_a-e_b+1}$ .

An  $f$ -commuting operator with this extra drop induces 0 on every page, because its image lies in the denominator. The induced operators therefore commute. Letting  $x_j, \xi_j \in T$  act by  $R_{x_j}$  and  $R_{\partial_j}$  makes every  $A_r, B_r$  a  $T$ -module and every  $d_r$   $T$ -linear. For  $r = 0$  this is the  $T$ -structure of  $M = \text{gr } Q$ , so  $A_1 \cong V$  and  $B_1 \cong U$  as  $T$ -modules. In addition,  $A_{r+1} \cong \ker d_r$  and  $B_{r+1} \cong \text{coker } d_r$  as  $T$ -modules. Indeed, if  $d_r([z]) = 0$ , write  $f(z) = f(w) + v$  with  $w \in G^{p+1}$  and  $v \in G^{p+r+1}$ . Then  $z - w$  represents the same source class and lies in  $G^p \cap f^{-1}G^{p+r+1}$ ; its kernel of passage back is exactly the denominator of  $A_{r+1}^p$ . For the target, quotienting  $B_r^{p+r}$  by  $d_r(A_r^p)$  enlarges the boundary denominator from  $G^{p+r} \cap f(G^{p+1})$  to  $G^{p+r} \cap f(G^p)$ , giving  $B_{r+1}^{p+r}$ . Consequently  $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$  is exact.

*Step 2: exhaustion, before localization.* Let  $y \in G^p$  represent a class of  $B_1^p$ . By Proposition 5.3,  $y = zx_1$  for one  $z \in Q$ , and by exhaustiveness  $z \in G^s$  for some  $s$ . For  $r \geq \max(1, p - s + 1)$  we have  $z \in G^{p-r+1}$ . Hence  $y \in G^p \cap f(G^{p-r+1})$ , and the class of  $y$  vanishes in  $B_r^p$ . In terms of order: if  $y \in F_m Q$  and the chosen preimage lies in  $F_{m'} Q$ , the class dies on page  $r = \max(1, m' - m + 1)$ . The maps  $\beta_r: B_1 \rightarrow B_r$ , the identity on representatives, are surjective and  $T$ -linear, and their kernels increase with  $r$ . An element of  $B_1$  has finitely many homogeneous components, so taking the largest  $r$  shows that every  $u \in B_1$  has some  $r$  with  $\beta_r(u) = 0$ .

*Step 3: localization.* Let  $S = T \setminus \mathfrak{q}$ . By Lemma 6.2,  $S^{-1}B_1 \cong U_{\mathfrak{q}}$  has finite length, so it is Noetherian over  $T_{\mathfrak{q}}$ . The maps  $S^{-1}\beta_r$  are surjective with increasing kernels. Each  $u/s$  is killed by  $S^{-1}\beta_r$  for the  $r$  belonging to  $u$ . The kernels therefore stabilize at all of  $S^{-1}B_1$ , and  $S^{-1}B_R = 0$  for some  $R$ .

*Step 4: telescoping.* Localizing the exact sequences  $0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0$ , all modules have finite length because  $A_1 \cong V$  and  $B_1 \cong U$  do, and subsequent pages are submodules and quotients. Additivity gives  $\ell(S^{-1}B_r) + \ell(S^{-1}A_{r+1}) = \ell(S^{-1}A_r) + \ell(S^{-1}B_{r+1})$ . Summing for  $r = 1, \dots, R-1$  cancels every intermediate term and gives  $\ell(U_{\mathfrak{q}}) + \ell(S^{-1}A_R) = \ell(V_{\mathfrak{q}}) + \ell(S^{-1}B_R) = \ell(V_{\mathfrak{q}})$ .

## Comparison COMP-07

**AI comment COMP-07. Information — proof comparison; no edit requested.** The formal proof uses Gabber directly for minimal primes. If such a prime contained  $x_1$ , the  $N$ -fold bracket of the monic symbol  $P$  with  $x_1$  would put the unit  $\pm N!$  in that prime. The subsequent stable-kernel length computation and Nakayama argument agree with the printed proof. [Mathematical proof \(COMP-07, p. 5\)](#).

### AI addition: Proof as formalized in Lean.

The length computation (stable kernel  $K = \ker x_1^m$ , the split  $\ell(U_{\mathfrak{q}}) = \ell(V_{\mathfrak{q}}) + \ell((M'/K)/x_1(M'/K))$ , and Nakayama) is formalized as printed. Only the first step differs: Lean does not pass through Theorem 3.8 and Lemma 3.9. Its formal Gabber theorem is stated directly for minimal primes: every minimal prime  $\mathfrak{p}$  of  $\text{Ann}_{k[x, \xi]} \text{gr } Q$  satisfies  $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$ . Suppose  $x_1 \in \mathfrak{p}$ . Since

$P \in \text{Ann gr } Q \subseteq \mathfrak{p}$  (Corollary 7.1), induction gives  $\{x_1, \{x_1, \dots, \{x_1, P\}\}\} \in \mathfrak{p}$  ( $j$  brackets) for every  $j$ . With the paper's bracket,  $\{x_1, g\} = -\partial g / \partial \xi_1$ . By Lemma 4.5,  $P$  is homogeneous of degree  $N$  with coefficient 1 at  $\xi_1^N$ . The  $N$ -fold bracket is therefore the constant  $(-1)^N \partial_{\xi_1}^N P = (-1)^N N!$ , which is a unit in characteristic zero, contradicting  $\mathfrak{p} \neq (1)$ . Lean's bracket is the negative of the paper's, so there the constant is  $N!$ . Lean then transfers this avoidance of  $x_1$  to the minimal primes of the annihilator of  $M' = M_{\mathfrak{q}}$ , using  $\text{Ann}_{C'} M' = S^{-1} \text{Ann}_C M$  for the finite module  $M$ .

### Comparison COMP-08

**AI comment COMP-08. Information — proof comparison; no edit requested.** The formal conclusion is scheme-theoretic:  $\text{Supp}(\text{gr } Q) \cap V(x_1) = \emptyset$ , equivalently  $\text{Ann}(\text{gr } Q) + (x_1) = (1)$ . It therefore persists under field extension. The formal statement uses the support transported by  $\xi \mapsto -\xi$ ; this fixes  $x_1$ , so the avoidance statements are equivalent. [Mathematical proof \(COMP-08, p. 5\)](#).

#### AI addition: Proof as formalized in Lean.

Lean states the conclusion scheme-theoretically:  $\text{Supp}_{k[x, \xi]}(\text{gr } Q) \cap V(x_1) = \emptyset$  in  $\text{Spec } k[x, \xi]$ , for an arbitrary field  $k$  of characteristic zero. For the finite module  $M = \text{gr } Q$  this is equivalent to  $\text{Ann}(\text{gr } Q) + (x_1) = (1)$ . It is an identity of ideals, so it persists under every field extension, and it gives the point-set statement over any algebraically closed extension. The proof is the printed one: Lemma 6.5 gives  $M/x_1 M = 0$ , and  $\text{Supp}(M/x_1 M) = \text{Supp } M \cap V(x_1)$  for finite  $M$ .

The formal statement is phrased for the transposed support, i.e. the support of  $\text{gr}$  of the left module obtained from  $Q$  through the transposition  $x_i \mapsto x_i$ ,  $\partial_i \mapsto -\partial_i$ . That support is the preimage of  $\text{Supp gr } Q$  under  $\xi \mapsto -\xi$ . This involution fixes  $x_1$  and hence  $V(x_1)$ , so the two disjointness statements are equivalent, and the formal consumer converts back at once to  $\text{Ann}(\text{gr } Q) + (x_1) = (1)$ . No left-module input is used anywhere.

### Comparison COMP-10

**AI comment COMP-10. Information — proof comparison; no edit requested.** The formal tangent-limit criterion is auxiliary. It assumes a generically smooth arc and a direct-summand lattice whose generic span is the lifted tangent space, with the specified axis component vanishing on its residue. Its proof does not use the smoothness hypothesis or the stated rank, but these remain assumptions of the formal statement. The formal proof of Theorem 8.2 follows a separate Laurent-series route, so it does not certify every step of the printed tangent-limit argument. [Mathematical proof \(COMP-10, p. 5\)](#). ROUTE-01 states the correspondence status.

#### AI addition: Formal status of Lemma 8.1.

Lean proves the following, over any algebraically closed field  $K$  with no characteristic assumption. Let  $\mathfrak{p} \subset K[x_1, \dots, x_n]$  be prime,  $q \in K[[t]]^{n+1}$  with  $q_0 \neq 0$ , and fix an axis  $1 \leq s \leq n$ . Assume:

- some  $q_i$  is a unit, and every homogenization  $f^h$  with  $f \in \mathfrak{p}$  satisfies  $f^h(q) = 0$  in  $K((t))$ ;
- $y = (q_i/q_0)_{i \geq 1}$  is a smooth point of  $\mathfrak{p} \otimes K((t))$ ;
- $L \subset K[[t]]^{n+1}$  is a direct summand of rank  $d + 1$ , with  $d$  arbitrary;
- the  $K((t))$ -span of  $L$  equals the preimage of the Zariski tangent space of  $\mathfrak{p} \otimes K((t))$  at  $y$  under the derivative of dehomogenization at  $q$ ;
- every  $w \in L$  has  $w_s(0) = 0$ .

Then  $[dx_s]$  lies in the Zariski closure of the smooth conormal directions of  $V(\mathfrak{p})$ , that is, in  $\text{Dir}(V(\mathfrak{p}))$ . The smoothness hypothesis is carried but not needed for the conclusion, and the rank  $d$  plays no role. This criterion is auxiliary. The proof of Theorem 8.2 does not use Lemma 8.1; it uses the valuation-theoretic argument given above after that theorem's proof.

### Comparison COMP-11

**AI comment COMP-11. Checked endpoint.** Over an algebraically closed field of characteristic zero, the theorem proves  $[dx_1] \in \text{Dir}(Y)$ . The constant-coordinate case is immediate. Otherwise one retained divisor valuation and its normalized projective column yield the residue-field tangent bound and the derivative columns used to construct a conormal covector with residue  $e_1$ . The equation-conormal closure gives the projective direction. The technical supplement gives the argument and states which steps of the printed proof remain unverified. COMP-11.

### AI addition: Proof of Theorem 8.2 as formalized in Lean.

Let  $K$  be algebraically closed of characteristic zero, let  $\mathfrak{p} \subset K[x_1, \dots, x_n]$  be prime, and suppose  $Y = V(\mathfrak{p})$  avoids  $x_1 = 0$ . Write  $R = K[x]/\mathfrak{p}$ ,  $F = \text{Frac}(R)$ , and  $\bar{x}_i$  for the image of  $x_i$ . If  $\bar{x}_1$  is algebraic over  $K$ , it is a nonzero constant; then  $x_1 - c \in \mathfrak{p}$  for some  $c \neq 0$ , and  $e_1$  is the gradient of this equation. So assume  $\bar{x}_1$  is transcendental.

The construction chooses a divisorial valuation ring  $V \subset F$  with  $\bar{x}_1 \in \mathfrak{m}_V$ , a uniformizer  $t$ , and residue field  $\kappa$ . Completing  $V$  gives a map into  $\kappa[[t]]$  that sends the chosen uniformizer to  $t$ . The same valuation witness supplies a normalized projective column  $q = (q_0, \dots, q_n)$ , with some chart entry  $q_j = 1$  and  $q_1 = q_0 \bar{x}_1$ . In the completion,

$$q_0 = t^a u_0, \quad q_1 = t^b u_1, \quad 0 < a < b, \quad u_0(0) \neq 0.$$

Dehomogenizing gives  $y = (q_1/q_0, \dots, q_n/q_0) \in \kappa((t))^n$ . The induced field map  $F \hookrightarrow \kappa((t))$  is injective, so for every ground-field polynomial  $f \in K[x_1, \dots, x_n]$ ,  $f(y) = 0$  exactly when  $f \in \mathfrak{p}$ . Thus  $y$  is the embedded generic point of the component, with its full equation kernel.

The residues of the normalized column coordinates generate  $\kappa$  up to an algebraic extension. Since the characteristic is zero, that extension is separable, and the differentials of those residues span  $\Omega_{\kappa/K}$ . Set  $d = \dim_{\kappa} \Omega_{\kappa/K}$ . Choose  $d$  residue coordinates whose differentials form a basis, and choose the dual  $K$ -derivations  $D_1, \dots, D_d : \kappa \rightarrow \kappa$ . Apply each derivation coefficientwise to  $q_i(t) = \sum_{r \geq 0} c_{ir} t^r$  and set  $Z_{ij}(t) = \sum_{r \geq 0} D_j(c_{ir}) t^r$ . On the chosen rows the

constant coefficient matrix of  $Z$  is the identity. Its corresponding power-series minor is therefore a unit.

The ratios  $q_i/q_0$  generate  $F$ , and the same visible differential frame bounds the component-field differentials. The exact sequence

$$\mathfrak{m}_V/\mathfrak{m}_V^2 \longrightarrow \kappa \otimes_V \Omega_{V/K} \longrightarrow \Omega_{\kappa/K} \longrightarrow 0$$

shows that the residue differential space needs  $d$  generators and the one-dimensional space  $\mathfrak{m}_V/\mathfrak{m}_V^2$  contributes at most the uniformizer direction. The finite visible differential module and Nakayama's lemma then give

$$\dim_F \Omega_{F/K} \leq d + 1.$$

The conormal sequence identifies  $\Omega_{F/K}$  with the quotient of  $F^n$  by the gradients of polynomials in  $\mathfrak{p}$ . After extending to  $\kappa((t))$ , its dual is the tangent space at  $y$ . The full equation kernel ensures these are exactly the equations of the component. Therefore

$$\dim_{\kappa((t))} T_y V(\mathfrak{p}_{\kappa((t))}) \leq \dim_{\kappa} \Omega_{\kappa/K} + 1.$$

Now let  $v = q'(t)$  be the uniformizer derivative. The columns  $Z_{\bullet j}$  and  $v$  live in projective coordinates; in the affine chart their quotient-rule images have entries

$$\delta_j y_i = \frac{Z_{ij} q_0 - q_i Z_{0j}}{q_0^2}, \quad \partial_t y_i = \frac{v_i q_0 - q_i v_0}{q_0^2} \quad (1 \leq i \leq n).$$

Differentiating the base equations shows that these images are tangent vectors at  $y$ . Since the selected minor of  $Z$  is a unit, solve its  $d$  selected equations to choose coefficients  $\lambda_j(t)$  for which  $v - Z\lambda$  vanishes on those rows. The row  $q_0$  has order  $a$ , and every entry of  $Z_{0\bullet}$  has order at least  $a$ . But  $q'_0 = at^{a-1}u_0 + t^a u'_0$  has nonzero coefficient  $au_0(0)$  in degree  $a-1$ . Characteristic zero therefore shows that the correction cannot cancel that coefficient. If  $c$  is the least order among the entries of  $v - Z\lambda$ , then  $c \leq a-1$ ; divide by  $t^c$  to obtain a primitive vector  $\tau$ . Because  $t$  is invertible in  $\kappa((t))$ , its dehomogenized image is tangent as well.

In the  $q_1$  row, both  $q'_1$  and the coefficientwise columns have order at least  $b-1$ . Since  $c \leq a-1 < b-1$ , the normalized entry  $\tau_1$  still has positive order, so  $\tau_1(0) = 0$ . The selected entries of  $\tau$  are zero by the choice of  $\lambda$ . Form the matrix whose columns are the projective position  $q$ , the  $d$  coefficientwise columns  $Z_{\bullet j}$ , and  $\tau$ . The chart row isolates  $q$ ; the selected minor isolates the  $Z$  columns; and primitivity gives a nonzero pivot for  $\tau$ . Its reduction modulo  $t$  has full column rank, hence the power-series matrix has a left inverse.

At  $t = 0$  the distinguished  $q_1$  row vanishes on every column:  $q_1(0) = 0$ , the  $Z$  entries have order at least  $b$ , and  $\tau_1(0) = 0$ . Thus the pure axis covector annihilates the reduced matrix. The left inverse lifts it to an exact power-series covector  $\ell(t)$  annihilating the whole matrix, with residue  $\ell(0)$  equal to that axis covector. The quotient-rule formulas show that each row column  $Z_{\bullet j}$  and  $\tau$  lifts a tangent vector at  $y$ , while  $q$  is the radial position vector. Thus the  $d+2$  independent columns lie in the affine cone over the projective

tangent plane at  $[q]$ . The affine tangent at  $y$  has dimension at most  $d + 1$  by the bound above, so this cone has dimension at most  $d + 2$ ; the independent columns therefore span it. Since  $q_0 \neq 0$  in  $\kappa((t))$ , dehomogenization identifies the cone modulo its radial line  $\kappa((t))q$  with the affine tangent at  $y$ . The remaining  $d + 1$  quotient-rule columns give a basis of that tangent space. By the full equation kernel, the affine tail of  $\ell(t)$  lies in the equation-conormal locus for  $\mathfrak{p}$  over  $\kappa((t))$ , and its residue is  $e_1$ .

Finally, a homogeneous polynomial vanishing on the smooth conormal directions vanishes on the equation-conormal locus: write a covector as a finite sum of gradients, use density of the smooth locus and the Nullstellensatz, then extend the resulting polynomial identity to  $\kappa((t))$ . Specializing along the power-series covector shows that the polynomial vanishes at  $e_1$ . Hence  $[dx_1] \in \text{Dir}(Y)$ .

This proves the endpoint by a generic-divisor route. It does not identify the retained valuation ring with the local ring of the divisor on the normalized projective closure, or construct the paper's selected smooth closed point, completed local coordinates, tilted arc, or tangent-frame specialization. Those steps of the printed proof remain a separate correspondence question.

### Comparison COMP-13

**AI comment COMP-13. Information — no further paper edit requested.** The formal proof uses the same finite Hamiltonian translations and component conormal containment. The equality of component ideals on a dense open set is checked through a separator and the product rule, then extended by closure. The limiting direction is used in the closure of fibre directions, as in the marked GEO-03 correction. [Mathematical proof \(COMP-13, p. 8\).](#)

#### AI addition: Proof of Theorem 9.2 as formalized in Lean.

Work in the setting of the formal statement and suppose  $W \cap \pi^{-1}(H) = \emptyset$ . Put  $J = I(W)$ , which is radical, and  $B = J \cap K[x]$ .

- Step 1.** ( *$\xi$ -homogeneity.*) Write  $g \in J$  as  $g = \sum_d g_d$  with  $g_d$  homogeneous of degree  $d$  in  $\xi$ . For  $(y, \xi) \in W$ , the polynomial  $a \mapsto \sum_d a^d g_d(y, \xi)$  vanishes on the infinite set  $K^\times$ . So every  $g_d \in J$ , i.e.  $J$  is  $\xi$ -homogeneous.
- Step 2.** (*Zero section.*) Let  $Y = V(B)$ . For  $y \in Y$  and  $g \in J$ ,  $g(y, 0) = g_0(y) = 0$  because  $g_0 \in J$  has  $\xi$ -degree 0, i.e.  $g_0 \in B$ . So  $Y \times \{0\} \subset V(J) = W$ . Hence  $\pi(W) = Y$  is closed, and  $Y \cap H = \emptyset$ .
- Step 3.** (*Vertical translation.*) Let  $f \in B$ . Then  $f \circ \pi \in J$ , so  $H_f g := \sum_i \partial_{x_i} f \partial_{\xi_i} g = \pm \{f, g\}$  lies in  $J$  for every  $g \in J$ . This is the only use of involutivity. Fix  $y \in Y$  and  $g \in J$  and let  $\varphi(t) = g(y, t df_y) \in K[t]$ . The chain rule gives  $\varphi^{(m)}(t) = (H_f^m g)(y, t df_y)$ , and  $H_f^m g \in J$ , so  $\varphi^{(m)}(0) = 0$  by Step 2. As  $m! \neq 0$  in  $K$ , we get  $\varphi = 0$ , so  $(y, t df_y) \in W$  for all  $t$ . Since  $\sum_a c_a d(f_a)_y = d(\sum_a c_a f_a)_y$  and  $\sum_a c_a f_a \in B$ , it follows that  $\{y\} \times \text{span}\{df_y : f \in B\} \subset W$ . Each translation is the finite polynomial Hamiltonian flow; the proof

composes these flows for the finite conormal decomposition, so no infinite exponential series is used.

**Step 4.** (*Component containment.*) Let  $\mathfrak{q}$  be a minimal prime of  $B$  and  $Y_0 = V(\mathfrak{q})$ . Taking a product of elements of the other minimal primes that lie outside  $\mathfrak{q}$  gives  $s \notin \mathfrak{q}$  with  $s\mathfrak{q} \subset B$ . For  $y \in Y_0$  with  $s(y) \neq 0$  and  $h \in \mathfrak{q}$ , we have  $d(sh)_y = s(y)dh_y$ , so  $(y, dh_y) \in W$  by Step 3. For  $g \in J$ , the polynomial  $G(x) = g(x, \nabla h(x))$  therefore vanishes on  $Y_0 \cap \{s \neq 0\}$ . Hence  $sG \in \mathfrak{q}$ , so  $G \in \mathfrak{q}$ , and  $(y, \nabla h(y)) \in W$  for every  $y \in Y_0$ . Absorbing linear combinations into  $h$  gives  $\{y\} \times \text{span}\{\nabla h(y) : h \in \mathfrak{q}\} \subset W$  for all  $y \in Y_0$ . In particular  $N_y^* Y_0 \subset W_y$  at every smooth point  $y$  of  $Y_0$ , which proves conormal containment.

**Step 5.** (*Conclusion in the affine cone.*)  $Y_0$  is irreducible, nonempty and disjoint from  $H$ . By Step 4 and  $W \subset V(P)$ ,  $P$  vanishes on  $N(Y_0) = \bigcup_{y \in (Y_0)_{\text{sm}}} N_y^* Y_0 \subset K^n$ . By part (i) of Theorem 8.2 in its formal version,  $P(1, 0, \dots, 0) = 0$ , a contradiction.

Part (i) is a statement about the vanishing ideal of the cone  $N(Y_0) \subset K^n$ . So  $[dx_1]$  never has to be a fibre direction of  $W$  over a finite base point, which removes the closure issue of GEO-03. Homogeneity of  $P$  is not used, and one component  $Y_0$  suffices.

## Comparison COMP-15

**AI comment COMP-15. Information — no further paper edit requested.** Lean constructs the quotient scalar-extension isomorphism, checks the associated-graded annihilator comparison, and descends quotient vanishing by faithful scalar extension. Geometry runs over  $\bar{k}$ ; rank-zero and scalar cases are handled separately. [Mathematical proof \(COMP-15, p. 9\).](#)

### AI addition: Proof as formalized in Lean.

Lean works over  $k$  and runs the geometry over  $\bar{k}$ . It checks the paper's quotient scalar-extension and faithful-descent route. The proof has six steps.

*Step 1: degenerate cases.* For  $n = 0$ ,  $A_0(k) = k$  and  $d \neq 0$  is a unit, so  $R = d^{-1}$  and  $F = S = 0$  work. Let  $n \geq 1$  and  $N = \deg_B d$ . If  $N = 0$ , then  $d = c \in k^\times$ , and  $\ell = x_1$ ,  $F = \ell^0 = 1$ ,  $R = c^{-1}$ ,  $S = 0$  work.

*Step 2: normal form.* For  $N \geq 1$ , the formal version of Proposition 4.2 gives symplectic  $M, M'$  with  $MM' = M'M = 1$  and  $c \in k^\times$  such that  $\tilde{d} = c^{-1}\Phi_M(d) \in A_n(k)$  has Bernstein degree at most  $N$  and coefficient 1 on  $p^N$ . Let  $I = \tilde{d}A_n + x_1^N \tilde{d}A_n$  and  $Q = A_n/I$ . It suffices to show  $Q = 0$ .

*Step 3: vanishing over  $\bar{k}$ .* The image of  $\tilde{d}$  in  $A_n(\bar{k})$  is again PBW-monic, so Sections 5–7 apply to  $Q_{\bar{k}} = A_n(\bar{k})/I_{\bar{k}}$ . Suppose  $J := \sqrt{\text{in}(I_{\bar{k}})} \neq (1)$ , and let  $W = V(J) \subseteq \bar{k}^{2n}$ . Then:

- $W \neq \emptyset$  by the Nullstellensatz, and  $I(W) = J$ .
- $W$  is closed and fibre-conical. The generators  $\sigma_L(z)$  of  $\text{in}(I_{\bar{k}})$  are homogeneous of degree  $L$  in  $\xi$ , so  $\text{in}(I_{\bar{k}})$  and  $J$  are  $\xi$ -homogeneous. For  $f \in J$  each  $\xi$ -homogeneous component  $f_j$  lies in  $J$ , and  $f(y, t\xi) = \sum_j t^j f_j(y, \xi)$ .

- $W$  is involutive:  $J = \sqrt{\text{Ann gr } Q_{\bar{k}}}$  is involutive by Theorem 3.8 (radical form, a corollary of the minimal-prime form).
- $P = \sigma(\tilde{d}) \in \text{in}(I_{\bar{k}}) \subseteq J$ , so  $W \subseteq V(P)$ , and  $P(1, 0, \dots, 0) = 1$  by Lemma 4.5.
- $W \cap \pi^{-1}(H) = \emptyset$  by Proposition 7.2 over  $\bar{k}$ , i.e.  $\text{in}(I_{\bar{k}}) + (x_1) = (1)$ .

The formal Theorem 9.2 gives a point of  $W$  with  $x_1 = 0$ , a contradiction. Hence  $\text{in}(I_{\bar{k}}) = (1)$ .

*Step 4: quotient and descent.* Lean constructs  $Q \otimes_k \bar{k} \cong A_n(\bar{k})/I_{\bar{k}}$  using the actual quotient and coefficient-extension maps. It also proves  $\text{Ann gr } Q_{\bar{k}} = (\text{Ann gr } Q)\bar{k}[x, \xi]$ . Empty characteristic support gives  $Q_{\bar{k}} = 0$  by the exhaustive filtration. Faithful scalar extension of vector spaces then gives  $Q = 0$ .

*Step 5: the certificate over  $k$ .* Since  $Q = 0$ , we have  $1 \in I$ . So  $1 = \tilde{d}R + x_1^N \tilde{d}S$  with  $R, S \in A_n(k)$ .

*Step 6: transport back.* Since  $c^{-1}$  is central, we get

$$1 = \Phi_M(d)(c^{-1}R) + x_1^N \Phi_M(d)(c^{-1}S).$$

Applying  $\Phi_M^{-1} = \Phi_{M'}$  gives  $1 = dR' + \ell^N dS'$  with  $R' = \Phi_{M'}(c^{-1}R)$ ,  $S' = \Phi_{M'}(c^{-1}S)$  and  $\ell = \Phi_{M'}(x_1) = \sum_j M'_{1j} z_j$ , where row 1 is the row of  $x_1$  and  $N = \deg_B d$ .

## Comparison COMP-16

### AI addition: Proof as formalized in Lean.

No two-generator theorem and no Ore condition are needed. Let  $T$  be a finitely generated torsion right module over a ring  $A$  without zero divisors in which every  $d \neq 0$  admits  $1 = dR + FdS$ ;  $A = A_n(k)$  qualifies by Theorem 1.1 and because  $A_n(k)$  is a domain (the Bernstein symbol of a product is the product of the symbols, in the polynomial ring  $k[x, \xi]$ ).

*The paper's matrix.* Let  $t_1, t_2 \in T$ . Choose  $d_2 \neq 0$  with  $t_2 d_2 = 0$  and  $e \neq 0$  with  $(t_1 d_2)e = 0$ ; then  $d := d_2 e \neq 0$  annihilates  $t_1$  and  $t_2$ . Write  $1 = dR + FdS$  and put  $a = dS$ . The two columns  $(a, 1 - Fa)$  and  $(1, -F)$  span  $A \oplus A$ : for  $(x, y)$  take  $u = y + Fx$  and  $v = x - a(y + Fx)$ . The first column lies in the kernel of  $(r, s) \mapsto t_1 r + t_2 s$ . Thus its image is generated by  $g = t_1 - t_2 F$ , and  $t_1 A + t_2 A = gA$ . This is the displayed matrix calculation in the paper, checked directly.

*Induction.* If  $t_1, \dots, t_m$  generate  $T$ , replace  $t_{m-1}, t_m$  by a single generator of  $t_{m-1}A + t_m A$  and repeat; after  $m - 1$  steps one generator remains.

## REFERENCES

GRAZ UNIVERSITY OF TECHNOLOGY